

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Pseudocode Design	Assessment:	Assessment Tool 1-Pseudocode Design
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	Develop and analyze machine learning models using linear regression, logistic regression, and unsupervised learning techniques for prediction, classification, model evaluation, and pattern discovery. (BL4)		
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Section / Batch:	CSE – AI	Date:	25/08/26

Code of Conduct

I V.Sai Charan – 192472159 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Sai Charan

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
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Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	C l e a r , w e l l - s t r u c t u r e d , a n d e a s y t o u n d e r s t a n d	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

QUESTION 11: LINEAR REGRESSION – MSE AND R² SCORE

Problem Statement

Linear Regression is a supervised machine learning technique used to predict a continuous numerical value from one or more input features. In this problem, a Linear Regression model is trained using a given dataset containing independent variables and a dependent variable. The dataset is divided into training data and testing data so that the performance of the trained model can be evaluated. The model learns the relationship between the input features and the target value from the training data. After training, the testing data is provided to the model to obtain predicted values. The difference between the actual and predicted values is called the residual. Mean Squared Error (MSE) is calculated to measure the average squared prediction error. Finally, the R² score is calculated to determine how well the model explains the variation in the target values.

Algorithm

1. Start the Linear Regression model evaluation process.
2. Load the dataset containing the independent variables and dependent target variable.
3. Identify and select the input features required for prediction.
4. Identify the target variable that the model has to predict.
5. Separate the input features and target values into X and Y.
6. Divide the dataset into training data and testing data.
7. Create a Linear Regression model for learning the relationship between X and Y.
8. Train the Linear Regression model using the training dataset.
9. Provide the testing input data to the trained model.
10. Calculate the predicted target values for the testing dataset.
11. Calculate the residual for each observation using actual value minus predicted value.
12. Calculate the Mean Squared Error by taking the average of squared residuals.
13. Calculate the R² score using the actual and predicted target values.

14. Display the predicted values and residual values obtained from the model.
15. Display the MSE and R^2 score as the final evaluation results.
16. Stop the process.

Pseudocode

BEGIN

LOAD dataset

SELECT input features

STORE as X

SELECT target variable

STORE as Y

SPLIT X and Y into

Training Data

Testing Data

CREATE Linear Regression model

TRAIN model using Training Data

PREDICT target values using Testing Data

STORE predicted values as Y_PRED

FOR each test observation

Residual = Actual Value – Predicted Value

END FOR

Calculate MSE

MSE = Average of

$(\text{Actual Value} - \text{Predicted Value})^2$

Calculate R^2 Score

DISPLAY Predicted Values

DISPLAY Residuals

DISPLAY MSE

DISPLAY R^2 Score

END

Sample Output

Linear Regression Model

Predicted Values:

[52.4, 61.8, 70.5, 78.2, 85.6]

Residuals:

[1.6, -1.8, 2.5, -2.2, 1.4]

Mean Squared Error (MSE): 4.05

R² Score: 0.94

Model evaluation completed successfully.

QUESTION 12: LOGISTIC REGRESSION – STUDENT PASS OR FAIL

Problem Statement

Logistic Regression is a supervised machine learning algorithm used mainly for classification problems. In this problem, Logistic Regression is used to predict whether a student will Pass or Fail based on study hours and attendance percentage. The student dataset contains study hours, attendance percentage, and the corresponding result of the student. Study hours and attendance percentage are considered input features, while Pass or Fail is considered the target variable. The dataset is divided into training and testing data to train and evaluate the classification model. After training, the model receives the study hours and attendance percentage of a new student. It calculates the probability that the student will pass. A threshold value of 0.5 is then used to convert the probability into the final Pass or Fail classification.

Algorithm

1. Start the student result prediction process.
2. Load the student dataset containing study hours, attendance percentage, and results.
3. Select study hours as the first input feature.
4. Select attendance percentage as the second input feature.
5. Select Pass or Fail as the target variable.
6. Separate the input features and target variable into X and Y.
7. Divide the dataset into training and testing data.
8. Create a Logistic Regression classification model.
9. Train the model using the available training data.
10. Take the study hours of a new student as input.
11. Take the attendance percentage of the new student as input.
12. Use the trained model to calculate the probability of the student passing.
13. Compare the calculated probability with the threshold value 0.5.
14. If the probability is greater than or equal to 0.5, classify the student as Pass.
15. If the probability is less than 0.5, classify the student as Fail.

Pseudocode

BEGIN

LOAD student dataset

SELECT Study Hours

SELECT Attendance Percentage

STORE both as X

SELECT Student Result

STORE as Y

SPLIT dataset into

Training Data

Testing Data

IF probability ≥ 0.5 THEN

Result = "PASS"

Result = "FAIL

END IF

DISPLAY probability

DISPLAY Result

END

Sample Output

Student Details

Study Hours: 6 hours

Attendance Percentage: 85%

Predicted Probability of Pass: 0.87

Predicted Result: PASS

Student classification completed successfully.

QUESTION 13: LOGISTIC REGRESSION – SIGMOID PROBABILITY

Problem Statement

Logistic Regression uses the sigmoid function to convert a numerical value into a probability between 0 and 1. In this problem, the input feature values, weights, and bias are used to calculate a linear value represented by z . The calculated z value is given as input to the sigmoid function. The sigmoid function produces a probability value that represents the likelihood of an observation belonging to class 1. A threshold value of 0.5 is used to convert the probability into a binary class. If the probability is greater than or equal to 0.5, the observation is classified as class 1. If the probability is less than 0.5, the observation is classified as class 0. Finally, the calculated probability and binary class are displayed as the output.

Algorithm

1. Start the sigmoid probability calculation process.
2. Input the required feature values for the classification problem.
3. Initialize the weights corresponding to the input features.
4. Initialize the bias value used in the logistic regression equation.
5. Calculate the linear value z using the input features, weights, and bias.
6. Apply the sigmoid function to the calculated z value.
7. Calculate the probability using the sigmoid formula.
8. Store the calculated value as the probability of class 1.
9. Compare the probability with the threshold value 0.5.
10. If the probability is greater than or equal to 0.5, assign class 1.
11. If the probability is less than 0.5, assign class 0.
12. Display the calculated sigmoid probability.
13. Display the final binary classification result.
14. Stop the process.

Pseudocode

BEGIN

INPUT feature value X

INPUT weight W

INPUT bias B

Calculate:

$$z = B + (W \times X)$$

Calculate sigmoid probability:

$$P = 1 / (1 + \text{EXP}(-z))$$

IF $P \geq 0.5$ THEN

Class = 1

ELSE

Class = 0

END IF

DISPLAY Probability P

DISPLAY Class

END

Sample Output

Input Feature Value: 2.5

Weight: 1.2

Bias: -1.0

Calculated z value: 2.0

Sigmoid Probability: 0.8808

Since Probability ≥ 0.5

Predicted Class: 1

QUESTION 14: LOGISTIC REGRESSION – EVALUATION METRICS

Problem Statement

A Logistic Regression classifier must be evaluated to determine how accurately it predicts the classes of new observations. In this problem, the actual class labels and predicted class labels are compared to create a confusion matrix. The confusion matrix contains four important values called True Positive, True Negative, False Positive, and False Negative. These values are used to calculate different performance measures for the classification model. Accuracy measures the overall percentage of correct predictions made by the classifier. Precision measures the correctness of positive predictions, while recall measures the ability to correctly identify actual positive cases. Finally, the F1-score combines precision and recall into a single performance measure. The calculated values are displayed to evaluate the overall performance of the Logistic Regression model.

Algorithm

1. Start the Logistic Regression evaluation process.
2. Obtain the actual class labels from the testing dataset.
3. Obtain the predicted class labels generated by the Logistic Regression model.
4. Initialize TP, TN, FP, and FN values to zero.
5. Compare each actual label with its corresponding predicted label.
6. If actual and predicted values are both 1, increase the True Positive count.
7. If actual and predicted values are both 0, increase the True Negative count.
8. If actual is 0 and predicted is 1, increase the False Positive count.
9. If actual is 1 and predicted is 0, increase the False Negative count.
10. Calculate accuracy using TP, TN, FP, and FN values.
11. Calculate precision using the True Positive and False Positive values.
12. Calculate recall using the True Positive and False Negative values.
13. Calculate F1-score using the calculated precision and recall values.
14. Display the confusion matrix values.

15. Display accuracy, precision, recall, and F1-score.
16. Stop the evaluation process.

Pseudocode

BEGIN

INPUT Actual_Labels

INPUT Predicted_Labels

SET TP = 0

SET TN = 0

SET FP = 0

SET FN = 0

FOR each observation

IF Actual = 1 AND Predicted = 1 THEN

TP = TP + 1

ELSE IF Actual = 0 AND Predicted = 0 THEN

TN = TN + 1

ELSE IF Actual = 0 AND Predicted = 1 THEN

FP = FP + 1

ELSE IF Actual = 1 AND Predicted = 0 THEN

FN = FN + 1

END IF

END FOR

Accuracy =

$(TP + TN) /$

$(TP + TN + FP + FN)$

Precision =

$TP / (TP + FP)$

Recall =

$$TP / (TP + FN)$$

F1_Score =

$$2 \times \text{Precision} \times \text{Recall} /$$
$$(\text{Precision} + \text{Recall})$$

DISPLAY TP

DISPLAY TN

DISPLAY FP

DISPLAY FN

DISPLAY Accuracy

DISPLAY Precision

DISPLAY Recall

DISPLAY F1_Score

END

Sample Output

Confusion Matrix Values

True Positive (TP): 45

True Negative (TN): 40

False Positive (FP): 5

False Negative (FN): 10

Accuracy: 0.85

Precision: 0.90

Recall: 0.82

F1-Score: 0.86

QUESTION 15: UNSUPERVISED LEARNING

Problem Statement

Unsupervised Learning is a machine learning approach in which the dataset does not contain predefined target or class labels. In this problem, an unlabelled dataset is provided to the learning system to discover hidden patterns and natural groups within the data. Since there is no target variable, the system works only with the available input features. First, the dataset is loaded and the required features are selected for analysis. The selected features are then preprocessed by checking missing values and scaling or normalizing the data when necessary. After preprocessing, an appropriate clustering algorithm is applied to identify similar groups of observations. Each observation is assigned to a suitable cluster based on its characteristics. Finally, the discovered clusters and cluster assignments are displayed as the output.

Algorithm

1. Start the unsupervised learning process.
2. Load the unlabelled dataset into the system for analysis.
3. Select the required input features from the dataset.
4. Check the selected features for missing or invalid values.
5. Handle missing values using an appropriate preprocessing method if required.
6. Convert the selected features into a suitable format for processing.
7. Scale or normalize the features when they have different ranges.
8. Select a suitable unsupervised learning or clustering algorithm.
9. Apply the selected algorithm to the preprocessed dataset.
10. Analyze the similarities and differences between the observations.
11. Identify the natural groups or clusters present in the dataset.
12. Assign each observation to its appropriate cluster.
13. Store the cluster number assigned to every observation.
14. Display the discovered clusters and their corresponding observations.
15. Verify that every observation has been assigned to a cluster.
16. Stop the unsupervised learning process.

Pseudocode

BEGIN

LOAD unlabelled dataset

SELECT required features

CHECK for missing values

IF missing values exist THEN

HANDLE missing values

END IF

PREPROCESS selected features

SCALE or NORMALIZE the features

SELECT clustering algorithm

APPLY clustering algorithm

to the preprocessed data

FOR each data point

ANALYZE data point

FIND appropriate cluster

ASSIGN data point

to the cluster

END FOR

STORE cluster assignments

DISPLAY discovered clusters

DISPLAY cluster assignment

for each data point

END

Sample Output

Unsupervised Learning Process

Number of Data Points: 10

Number of Features: 2

Preprocessing completed successfully.

Clusters Discovered: 3

Cluster Assignments:

Data Point 1 → Cluster 1

Data Point 2 → Cluster 1

Data Point 3 → Cluster 2

Data Point 4 → Cluster 2

Data Point 5 → Cluster 3

Data Point 6 → Cluster 1

Data Point 7 → Cluster 3

Data Point 8 → Cluster 2

Data Point 9 → Cluster 3

Data Point 10 → Cluster 1

Clustering completed successfully.